RICHA JHA

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PROFESSIONAL SUMMARY

Computer Science master's student at NC State with 3 years of software and data engineering experience at Deloitte and HighRadius, working across Python, Java, SQL, React, and AI/ML. Building AI applications with LLMs, RAG, and computer vision through technical projects and certifications like IBM AI Developer and Google's Generative AI Leader, while also building Product Management skills through cross-functional experience and a Microsoft AI Product Manager certification.

EDUCATION

North Carolina State University, Master of Computer Science *Anticipated May 2027*
Raleigh, NC | GPA: 3.61/4

Kalinga Institute of Industrial Technology (KIIT), B.Tech in Information Technology *Jul 2018 – May 2022*
India | CGPA: 9.07/10

PROFESSIONAL EXPERIENCE

Software Engineer, Deloitte *Jun 2022 – Jul 2025*

- Designed and automated ETL pipelines using Python, PySpark, SQL, Databricks, Informatica, and AWS, cutting manual effort by more than 400 hours a year and improving data quality and timeliness.
- Led client discussions and cross-team conversations with in-house and client stakeholders to prioritize the backlog, run Agile sprints, and communicate data risks and dependencies.
- Mentored new team members on coding practices, documentation, and quality standards.

Software Engineer Intern, HighRadius *Jan 2021 – May 2022*

- Built an AI-enabled B2B accounts receivable application end to end, automating invoice data and cutting collection cycle time by 45%.
- Implemented backend services in Java and SQL, and built frontend features in React.js for invoice processing workflows.

Backend Developer Intern, PrepEase *Apr 2020 – May 2020*

- Built backend APIs using Node.js and MongoDB for an online examination platform supporting 150+ concurrent users.
- Added features such as question uploads, test cards, countdown timers, and live video class integration.

PROJECTS

PulseMorph: AI Wearable Health Dashboard, WolfHacks Hackathon | GitHub *2026*
Contributed to the data pipeline feeding glucose, heart rate, and skin conductance data through Databricks into Tiger Data (PostgreSQL and TimescaleDB), and supported the Gemini API integration behind the app's real-time glucose predictions and AI agent, on a multi-person hackathon team.

Satellite Image Classification: Vegetation vs Built-Up | GitHub *2026*
Designed, trained, and evaluated a CNN with batch normalization on 16,000 EuroSAT satellite images to classify vegetation versus built-up land, reaching 93.3% test accuracy as lead contributor on a 3-person team.

SafeBites: AI Allergen Detection & Food Search Platform | GitHub *2025*
Built backend REST APIs (FastAPI, MongoDB) for an AI-powered food search platform with a 4-person team, integrating LangGraph, LangChain, FAISS, and RAG-based semantic search to flag allergens in restaurant menus.

Micro-Learning Platform, NC State WiCS × Fidelity Investments | GitHub *2026*
Selected from around 500 applicants for this Fidelity Investments-sponsored micro-learning platform team. Built Java backend REST APIs (Spring Boot, PostgreSQL) with JWT authentication, using a modular architecture and a team GitHub workflow.

AI B2B Invoice Management Application *2021*
Built an end-to-end ML model to predict invoice payment dates from historical invoice and payment data, achieving about 87% accuracy on a Java-SQL backend.

IoT Analytics: Weather Time-Series Forecasting | GitHub *2026*
Built a data pipeline on 4 years of NOAA weather data and benchmarked forecasting models (AR, Exponential Smoothing) against ML models (ANN, DBSCAN, HMM); ANN performed best with an RMSE around 2.2°F.

SKILLS

Languages: Python, Java, C++, C, JavaScript, SQL
AI, ML & Data Science: Machine Learning, LLMs, RAG, Statistics, Data Science, Artificial Intelligence, Data Analysis, Data Visualization, Power BI
Databases & Data Engineering: MongoDB, PostgreSQL, PySpark, Azure Databricks, ETL Development, Data Modeling, Database Management
Web Development: React.js, Node.js, HTML, CSS, Bootstrap, jQuery
Tools & Practices: Git, AWS, API Development, Software Development, Agile Methodologies, Human-Computer Interaction, Microsoft Excel

CERTIFICATIONS

- Google Generative AI Leader Professional Certificate, 2026 (generative AI strategy, prompt engineering, AI agents, responsible AI, Google Cloud)
- IBM AI Developer Professional Certificate, 2026 (Python, Flask, generative AI, LLMs, RAG)
- Deloitte AI Academy, 2022 (SQL, Power BI, Python, Azure, PySpark)
- Microsoft AI Product Manager Professional Certificate, 2026 (product strategy, roadmapping, market research, UX/UI design)

ACHIEVEMENTS

- Deloitte Spot Award (2022) and Deloitte Applause Award (2024) for outstanding contributions to the firm.
- Won Most Voted Software Engineering Project (2025) for SafeBites, and mentored new engineers at Deloitte.